

Experiment No.	02
Title	Design Webpage using HTML5 and CSS3
Course Outcome	CO1

1. Objective / Aim

To design and implement a modern, styled webpage for ANITS using HTML5 semantic elements and CSS3 features — including Flexbox layout, gradients, box shadows, border radius, and hover transitions — demonstrating the separation of structure (HTML) and presentation (CSS).

2. System Requirements

Hardware

Processor	Intel Core i5 or equivalent
RAM	8 GB or above
Storage	256 GB SSD / HDD

Software

Operating System	Windows 11 Home
Code Editor	Visual Studio Code with Live Server extension
Browser	Google Chrome / Microsoft Edge
Runtime	None — static HTML + CSS files

3. Architecture / Design

File Structure

folder structure

```
lab2_exp2/
├── lab2.html    <-- HTML5 structure (linked via <link rel="stylesheet">)
└── lab2.css     <-- CSS3 styling
```

Page Layout (HTML5 Structure)

- <header> — Flexbox layout: logo (h1) left, <nav> links right
- <section id="home"> — gradient hero banner with heading, paragraph, CTA link
- <div class="container"> [Flexbox row]
- → <main id="articles"> — CSE department article + Student Enquiry <form> (fieldset/legend, text, email, select, textarea, submit & reset)
- → <aside id="about"> — About ANITS, Key Highlights list, Departments Offered list
- <footer> — copyright text

CSS3 Features Demonstrated

- Flexbox Layout — header, .container, form inputs
- Linear Gradient — hero section background

- Box Shadow — main content card
- Border Radius — cards, buttons, form fields
- Transitions — nav link hover, button hover effects
- Pseudo-class styling — :hover on nav links and buttons

4. Source Code — lab2.html

lab2.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>ANITS - Anil Neerukonda Institute of Technology &
Sciences</title>
  <link rel="stylesheet" href="lab2.css">
</head>
<body>

  <!-- Header Section -->
  <header>
    <h1>ANITS</h1>
    <nav>
      <a href="#home">Home</a>
      <a href="#articles">Departments</a>
      <a href="#about">About</a>
      <a href="#contact">Contact Us</a>
    </nav>
  </header>

  <!-- Hero / Showcase Section -->
  <section id="home">
    <h2>Anil Neerukonda Institute of Technology & Sciences</h2>
    <p>A premier engineering institution in Visakhapatnam, focused on
academic excellence, research, and strong industry-oriented learning.</p>
    <p><a href="#articles"><b>Explore Our Departments Below
&darr;</b></a></p>
  </section>

  <!-- Main Content Layout -->
  <div class="container">

    <!-- Left Side: Main Content Column -->
    <main id="articles">

      <article>
        <h2>1. Computer Science Engineering</h2>
```

```
<p>Established with modern labs and industry-aligned
curriculum</p>

<p>The CSE department focuses on core programming, data
structures, AI/ML, and software development, preparing students for top
tech careers through strong placement support.</p>

<p>Students get hands-on exposure to real-world projects
using tools like <code>Python</code>, <code>Java</code>, and cloud
platforms.</p>
</article>

<article>
  <h2>2. Student Enquiry Form</h2>
  <p>For admissions and general queries</p>
  <p>Prospective students and parents can use the form below
to reach out to the ANITS admissions office for any information regarding
courses, fees, or campus facilities.</p>
  <br>
  <h3>Contact ANITS</h3>
  <form id="contact" action="#" method="post">
    <fieldset>
      <legend><b>Admission Enquiry Form</b></legend>
      <p>
        <label for="name">Full Name:</label>
        <input type="text" id="name" name="username"
required placeholder="John Doe">
      </p>
      <p>
        <label for="email">Email Address:</label>
        <input type="email" id="email" name="useremail"
required>
      </p>
      <p>
        <label for="topic">Enquiry Topic:</label>
        <select id="topic" name="usertopic">
          <option
value="admissions">Admissions</option>
          <option value="academics">Academic
Curriculum</option>
          <option
value="placements">Placements</option>
        </select>
      </p>
      <p>
        <label for="message">Message
Details:</label><br>
        <textarea id="message" name="usermessage"
rows="5" cols="40" placeholder="Type your query here..."></textarea>
      </p>
    </fieldset>
  </form>
</article>
```

```
<p>
    <input type="submit" value="Submit Enquiry">
    <input type="reset" value="Clear Entries">
</p>
</fieldset>
</form>
</article>

</main>

<!-- Right Side: Sidebar Column -->
<aside id="about">
    <h3>About ANITS</h3>
    <p>ANITS is known for its strong academic reputation, modern
infrastructure, and excellent placement record across top companies.</p>

    <h4>Key Highlights:</h4>
    <ul>
        <li>Experienced faculty</li>
        <li>Well-equipped labs</li>
        <li>Strong placement cell</li>
    </ul>

    <h4>Departments Offered:</h4>
    <ol>
        <li>Computer Science Engineering</li>
        <li>Electronics & Communication</li>
        <li>Mechanical Engineering</li>
    </ol>
</aside>

</div>

<!-- Footer Section -->
<footer>
    <p>&copy; 2026 ANITS - Anil Neerukonda Institute of Technology &
Sciences. All Rights Reserved.</p>
</footer>

</body>
</html>
```

4b. Source Code — lab2.css

lab2.css

```
/* ===== Reset & Base ===== */
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: 'Segoe UI', Arial, sans-serif;
  background-color: #f9f9f9;
  color: #333;
  line-height: 1.6;
}

/* ===== Header ===== */
header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  background-color: #1e3c72;
  color: white;
  padding: 15px 30px;
}

header h1 {
  font-size: 24px;
}

nav a {
  color: white;
  text-decoration: none;
  margin-left: 20px;
  font-weight: bold;
  transition: 0.3s;
}

nav a:hover {
  color: #ffcc00;
}

/* ===== Hero Section ===== */
#home {
  text-align: center;
  padding: 50px 20px;
```

```
background: linear-gradient(to right, #2a5298, #4facfe);
color: white;
}

#home h2 {
  font-size: 28px;
  margin-bottom: 15px;
}

#home p {
  max-width: 600px;
  margin: 0 auto 10px auto;
}

#home a {
  color: #ffcc00;
  text-decoration: none;
}

/* ===== Main Layout ===== */
.container {
  display: flex;
  gap: 25px;
  padding: 30px;
  align-items: flex-start;
}

main {
  flex: 7;
  background-color: white;
  padding: 25px;
  border-radius: 10px;
  box-shadow: 2px 2px 10px rgba(0,0,0,0.1);
}

aside {
  flex: 3;
  background-color: #f4f4f4;
  padding: 20px;
  border-radius: 10px;
}

/* ===== Articles ===== */
article {
  margin-bottom: 25px;
```

```
padding-bottom: 20px;
border-bottom: 1px solid #ddd;
}
```

```
article h2 {
    color: #1e3c72;
    margin-bottom: 8px;
}
```

```
article p {
    margin-bottom: 10px;
}
```

```
code {
    background-color: #eee;
    padding: 2px 6px;
    border-radius: 4px;
    font-family: monospace;
}
```

```
/* ===== Form ===== */
form fieldset {
    border: 1px solid #ccc;
    border-radius: 8px;
    padding: 20px;
}
```

```
form legend {
    padding: 0 10px;
    color: #1e3c72;
}
```

```
form label {
    display: inline-block;
    width: 130px;
    font-weight: bold;
}
```

```
form input[type="text"],
form input[type="email"],
form select,
form textarea {
    width: 60%;
    padding: 8px;
    margin: 6px 0;
```

```
border: 1px solid #ccc;
border-radius: 6px;
}

form input[type="submit"],
form input[type="reset"] {
padding: 8px 18px;
margin-top: 10px;
margin-right: 10px;
border: none;
border-radius: 6px;
background-color: #1e3c72;
color: white;
cursor: pointer;
}

form input[type="submit"]:hover,
form input[type="reset"]:hover {
background-color: #4facfe;
}

/* ===== Sidebar ===== */
aside h3 {
color: #1e3c72;
margin-bottom: 10px;
}

aside h4 {
margin-top: 15px;
margin-bottom: 8px;
}

aside ul, aside ol {
margin-left: 20px;
}

/* ===== Footer ===== */
footer {
text-align: center;
padding: 18px;
background-color: #1e3c72;
color: white;
margin-top: 20px;
}
```


5. Compilation / Execution Steps

- Step 1 Create a project folder named lab2_exp2/.
- Step 2 Create lab2.html inside the folder and paste the HTML source code.
- Step 3 Create lab2.css in the same folder and paste the CSS source code.
- Step 4 Open the lab2_exp2/ folder in Visual Studio Code.
- Step 5 Install the Live Server extension if not already installed.
- Step 6 Right-click lab2.html → select "Open with Live Server".
- Step 7 Browser opens at http://127.0.0.1:5500/lab2_exp2/lab2.html.
- Step 8 Verify all sections — header, hero, articles, form, sidebar, footer — render with full CSS3 styling.

6. Expected Output / Screenshot

The screenshot below shows the ANITS webpage rendered in a Chromium browser, demonstrating HTML5 semantic structure combined with CSS3 styling — gradient hero section, Flexbox layout, styled sidebar, CSS3 form, and a styled footer.

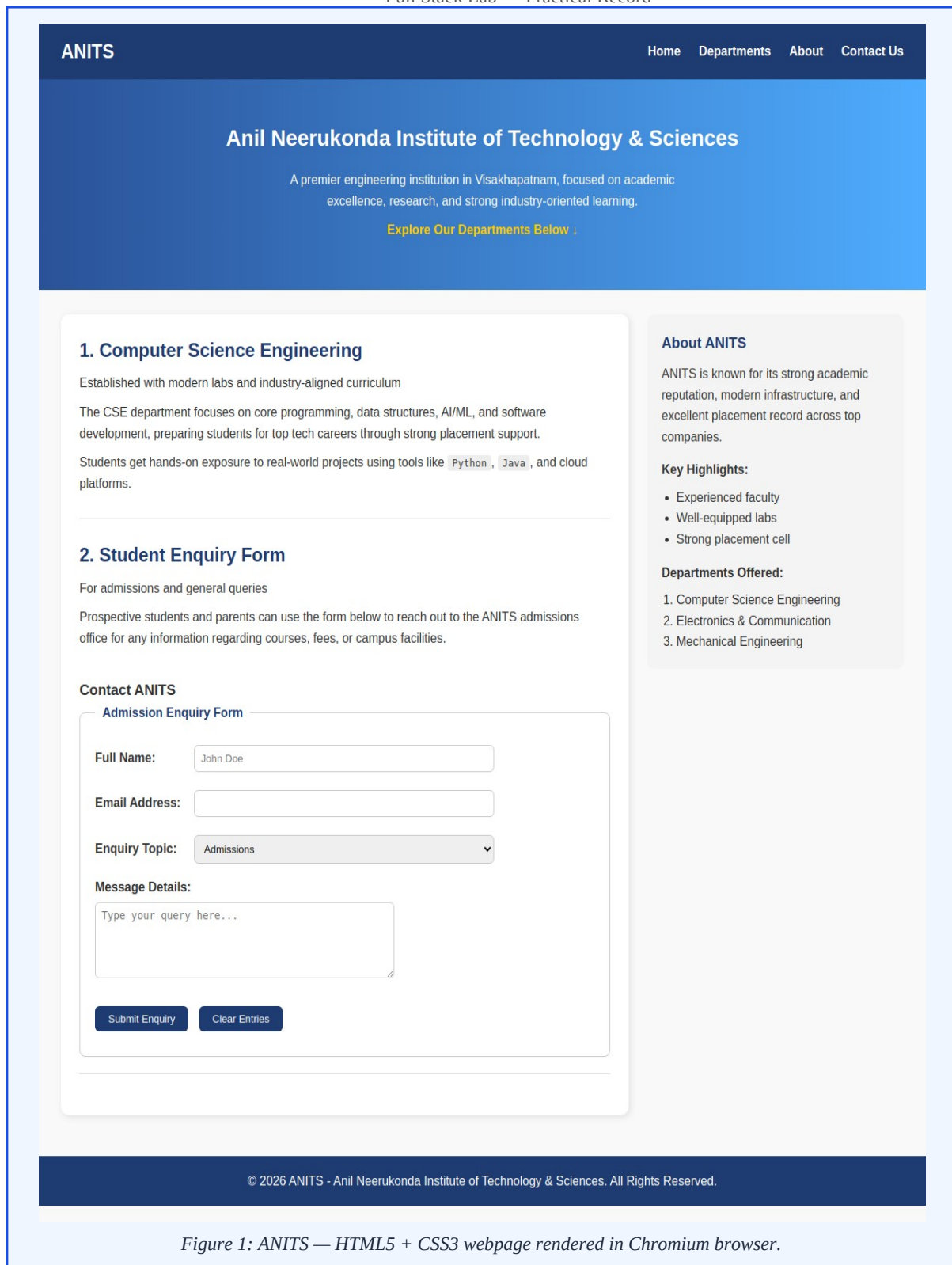


Figure 1: ANITS — HTML5 + CSS3 webpage rendered in Chromium browser.

7. Result / Conclusion

The experiment was successfully completed. A fully styled webpage was designed and implemented using HTML5 semantic elements — `<header>`, `<nav>`, `<section>`, `<main>`, `<article>`, `<aside>`, `<form>`, and `<footer>` — structured in `lab2.html`, with all visual styling defined in a separate `lab2.css` file using CSS3 features including Flexbox, linear gradients, box shadows, border

radius, and transitions. The separation of structure and presentation was clearly demonstrated, fulfilling Course Outcome CO1.

Student Signature

Faculty / Lab In-charge Signature